

0 4

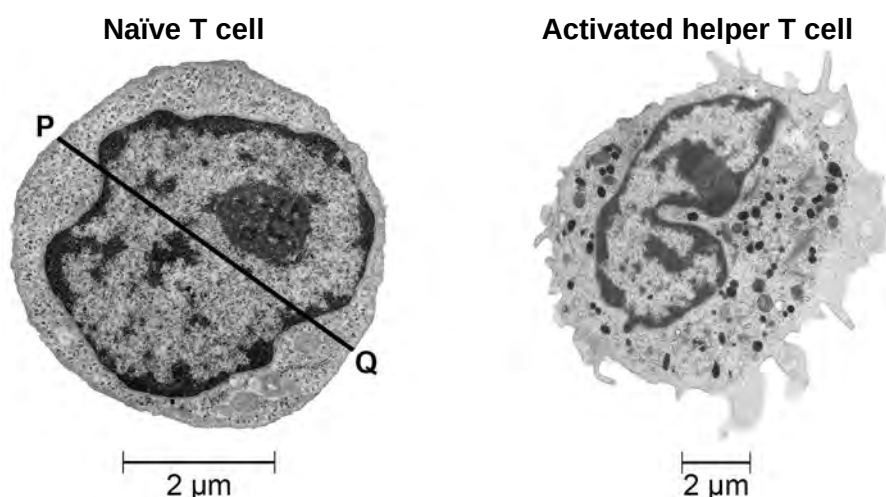
Scientists investigated the activation of T lymphocytes.

The scientists studied two types of cell:

- naïve T cells, which are T cells that have not yet been in contact with a foreign antigen
- activated helper T cells, which are T cells that have been activated by a foreign antigen.

Figure 5 shows electron microscope images of the two types of cell.

Figure 5



The activated helper T cell has a volume of $463 \mu\text{m}^3$

0 4 . 1

Calculate the volume of the naïve T cell shown in **Figure 5**.

Then calculate how many times larger the activated helper T cell volume is compared with the naïve T cell volume.

Assume the cell is spherical.

Use line **PQ** to measure the diameter of the naïve T cell.

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3 \text{ where } \pi \text{ is } 3.14$$

Show your working.

[3 marks]

Volume of naïve T cell _____ μm^3

Number of times larger the activated helper T cell
volume is compared with the naïve T cell volume _____



0 4 . 2

State **one** feature that shows the images in **Figure 5** were taken with an electron microscope and **not** an optical microscope.

Explain your answer.

[2 marks]

0 4 . 3

State **one** role of a helper T cell.

[1 mark]

0 4 . 4

The rate of translation is increased in T cells activated by antigens.

Describe the role of tRNA in translation.

[2 marks]

8

Turn over for the next question

Turn over ►

